



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 20-401-1

Date: 01-02-2021

EDMS 20-401-1

TECHNICAL SPECIFICATION

FOR

MEDIUM VOLTAGE SMART METER

Issue: FEB-2021 / Rev- 1



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Smart Meters- Three Phase In-Direct Meter

1. Smart Meters - General Requirements

Meter that connected through current and voltage transformer at medium voltage in aim to measure the total energy consumed from utility Distribution Company at certain point

1.1. Service Conditions

The meters to be supplied according to this specification shall be suitable for satisfactory continuous operation under the following environmental conditions:

Table (1)

NO	Condition	Specification
1	Maximum ambient temperature ($^{\circ}$ C)	70
2	Minimum ambient temperature ($^{\circ}$ C)	-5
3	Maximum daily average temperature ($^{\circ}$ C)	35
4	Maximum annual average temperature ($^{\circ}$ C)	30

1.2. Environmental Requirements

The meters shall conform to the environmental requirements as defined in IEC 62052-11, 62053-21 and 62053-23. In addition to those requirements, the following minimum requirements shall be met:

1. The meter shall be protected against malfunction due to the ingress of vermin, dust, or excess humidity by conformal coating of the printed circuit boards in the meter.
2. Any openings shall be as small as practically possible to prevent the ingress of dirt and vermin and to limit the potential for vandalism or tamper following Meter case.

1.3. Standards and codes of practice

All material and equipment supplied and all works carried out shall comply in every respect with the technical codes of the International Organization for Standardization (ISO) and with the recommendations of the International Electro technical Commission (IEC) and the European Standards (EN), which apply to the electrical equipment.

The following documents shall be read in conjunction with this specification. In case of conflict, however, this document shall take precedence. Nothing in this specification shall lessen the bidders obligations detailed in any other documents forming part of the contract.



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Table (2)

Identifier	Description
IEC 62052-11:2003	<i>Electricity metering equipment (a.c.) – General requirements, tests and test conditions – Part 11: Metering equipment</i>
BS EN50470-1	<i>Electricity metering equipment (a.c.) – General requirements, tests and test conditions – Part 1: Metering equipment</i>
IEC 62055	<i>Part 21: Framework for standardization Part 31: Particular requirements- Static payment meters for active energy (classes 1).</i>
BS EN50470-3	<i>Electricity metering equipment (a.c.) - Part 3: Particular requirements – Static meters for active energy (class indexes A, B and C)</i>
BS EN 61268	<i>Alternating Current Static Var-Hour Meters for Reactive Energy (Classes 2 and 3)</i>
IEC 62053-21:2003	<i>Electricity metering equipment (a.c) static watt hour meter for active energy meters class 1 & 2</i>
IEC62053-22:2003	<i>Electricity metering equipment (a.c.) – Particular requirements – Part 22: Static meters for active energy (classes 0,2 S and 0,5 S)</i>
IEC 62053-23	<i>Electricity metering equipment (a.c) static watt hour meter for reactive energy meters class 2 & 3</i>
IEC 60060	<i>HV test technique</i>
IEC 60068	<i>Shock and vibration withstand</i>
IEC 60387	<i>Marking symbols standard</i>
IEC 60529	<i>Degree of protection</i>
IEC 62054	<i>Real Time Clock (RTC)</i>
IEC 62056- 21:2002	<i>Electricity metering data exchange for meter reading tariff and load control : Direct local data exchange</i>
IEC 62056-51:1998	<i>Electricity metering data exchange for meter reading tariff and load control : Application layer protocols</i>
IEC62056-61	<i>Electricity metering – Data exchange for meter reading, tariff and load control –Part 61: Object identification system (OBIS)</i>



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1.4. General Electrical Requirements

The meters shall comply with the following electrical requirements:

Table (3)

No	Description	Specification
1	Rated Voltage	Medium Voltage MV Network Three Phase Meters: 3x63.5/110V 3x57/100V
2	Variation in voltage	+30% to -40%
3	Continuous over voltage	Medium voltage meters shall withstand 110 volts (Phase to neutral) for 48 hours without being damaged or causing changes of more than 0.01 kWh in its credit registers (excluding the possible decrement of credit due to power being consumed).
4	Rated frequency	50 Hz
5	Variation in frequency	± 5%
6	Power factor	lag – unity – lead
7	Power consumption in the voltage circuit	10VA (2W) for voltage circuit. 1VA for phase current circuit for transformer operated meters Note: the above value for power consumption is for meter without communication modem.
8	RTC	Built-in real time clock with 10 years of operation battery backup
9	Display	LCD; Visibility shall be sufficient to read the meter mounted at height of 0.5 to 2.0 m. Pin type; trans-reflective HTN or STN type industrial grade; temperature range – 20°C to +70°C Meter LCD Messages shall be available in Arabic language.
10	Digits	WxH: 4 mm x 8 mm (minimum) with backlight.
11	Maximum viewing angle	160 degrees
12	Memory	Nonvolatile memory (EEPROM) that retains information up to 20 years without being damaged or corrupted during the meter operation.
13	Lightning Protection	In accordance to IEC 1036:1990



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1.4.1. General Mechanical Requirements

The meters shall comply with the following mechanical requirements:

Table (4)

No	Description	Specification
1	Meter Case	The meter body shall be a nonflammable and high impact, strong material. The meter base, body, and frame including terminal block shall be of heat resistive; shock proof and rust proof good quality hard material (Compatible with IEC 695-2-1). The meter base will be provided with 3 (three) screw mounting holes, 1 (one) slotted meter support bracket at the top and 1 (one) round hole on each side in the bottom half of the base for securely mounting the meter to the meter board. The meter cover shall be made of polycarbonate material. The meter cover shall be provided with a window of polycarbonate or toughened glass. The meter shall be effectively sealed to prevent entrance of moisture, rain and dust into its internal part
2	Connections diagrams and terminal marking	Every meter shall be indelibly marked (engraved on the terminal cover surface) with a connection diagram. The meter terminals shall be marked and indicated on this diagram
3	Protection against penetration of dust and water	Conform to the degree of protection IP54 for outdoor placement.
4	Pulse output	Flashing LED visible from the front along with Auxiliary 2-wire terminal for : - Active and Reactive energy consumption rate
5	Temperature range	Limit range of operation: -25°C to 70°C Limit range of storage and transport: -25°C to 70°C
6	Name plate	Every meter shall have clearly visible, indelibly and distinctly marked name plate containing the following information: - Manufacturer's name - Meter type - Number of phases and number of wire - Meter serial number - year of manufacture - Rated voltage of the system - Basic current and maximum current - Reference frequency in hertz - Meter constant in imp/kWh - Meter Constant in imp/kVARh - Class index of the meter - DISCO logo - Contract No. - Reference Standards



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1.5. General Functional Requirements

1.5.1. Meter Sealing

1. Provision shall be for sealing the meter with at least two Ferrule stainless steel seals
2. The stainless steel seals shall be applied in such a way that it will not be possible to undo/loosen the mounting screws used to secure the meter without breaking these seals.
3. The stainless steel seals shall be applied in such a way that they will be easily visible when viewing an installed meter from the front.
4. The terminals shall also be having Ferrule/TT sealing arrangement.

1.5.2. Smart Meter RS-485 Serial Communication Interface

In this specification, RS-485 serial communication interface is used for physical layer Communication interface standard.

1.5.3. Meter Local Communication Interface

1. The meter shall have an IEC 62056-21 compliant optical communication port. This shall be place to allow the Utility to access via the front of the meter for a variety of interactions via a hand held unit (HHU).
2. The meter shall have a 2-wire (+Ground Reference) serial communication port
3. The meter shall have an RS485 compatible with IEC 62056 communication protocol for a local connected network of 30 meters (without a repeater) or Local connection with nearest data concentrator (RS gateway) in the field of installation.
With one wide area network (WAN) Gateway.

1.5.4. Meter Remote Communication Interface

- The meter shall have remote communication option which will be able to interface with RS-485, RJ-45 port for data communication with the central server from meters via one of the different communication media (RS-485 Serial Communication as the primary communication type).
- Communication device shall be a modem using GSM and 4G mobile network communication. In addition, or in replacement, it should be possible to connect an external modem (PSTN, GSM, 4G)
- The communication modules works as a plug-and-play model, without the need to modify/change the meter firmware.
- The communication protocol shall be compatible with the IEC 62056 Dynamic DLMS/COSEM Application Layer Protocol. DLMS Compliance Certificate (mandatory) ,The meter shall have a certificate for IEC62056 DLMS/COSEM, for the following:
 - Optical port Communication
 - 2-wire Serial RS485 Communication port
 - RS-485 COMMUNICATION SPECIFICATION



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- RS-485, also known as TIA-485(-A), EIA-485, is a standard defining the electrical characteristics of drivers and receivers for use in serial communications systems. Electrical signaling is balanced, and multipoint systems are supported.
- The standard is jointly published by the Telecommunications Industry Association and Electronic Industries Alliance (TIA/EIA). Digital communications networks implementing the standard can be used effectively over long distances and in electrically noisy environments.

Table (5)

Item	Specification
Transmission Mode	Half Duplex
Wiring	2 Wires + Ground Reference
Cable Twisting	Twisted Pair
Slew Rate	Limited
Baud Rate	115200bps
Topology	Bus
Connector Type	RJ45 or Wiring Terminals
Cable Type	CAT 5e or CAT 6
Cable Shielding	Unshielded or Shielded
Ground Shield	Single point
Cable Characteristic Impedance	120Ω
Wires Labelling	Color coding to be specified
Terminating Resistors	2 Ends
Terminating Resistor Value	120Ω
Flow Control	None 1 Stop bit
Parity	None
Fail Safe Bias	Pull up A Pull down B



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RS-485 Gateway Specifications (If required)

Table (6)

Specification
2 RS-485 Modules (Supporting up to 60 nodes) Asynchronous up to 2Mbps RTS/CTS and SCADA flow control (Modbus, IEC-101/102, IEC-104 gateway)
Uplink LTE Module
2 SIM Trays with Dual SIM Redundancy (internal)
2 SMA antennas
1 Ethernet Port (RJ45 connector)
Multiple power supply options (DC, AC, Multirange)
Temperature Range -20 to 70 °C
CLI configuration and Management
Security: IPSec support in transport and tunnel mode Pre-shared authentication, RSA, Certificates, MD5, SHA-1 DES (56 bits), 3DES (168 bits), AES (128, 192 and 256 bits), IKEv1, IKEv2
Certified for use in electrical environments The routers are certified for use in electrical environments with EMC, isolation, immunity, electrical, climatic and mechanical certificates. Certifications: Isolation (EN 60255-5) for electrical resistance and impulses Climatic: EN 60068-2-78, EN 60068-2-2, EN 60068-2-1, EN 60068-2-14 Mechanical: EN 60870-2-2, EN 60068-2-6 EN 60068-2-27 Immunity (I): EN 61000-4-2, EN 61000-4-3, EN 61000-4-4, EN 61000-4-5 EN 61000-4-6, EN 61000-4-8, EN 61000-4-10, EN 61000-4-12, EN 61000-4-13 Immunity (II): EN 61000-4-18 Electrical: EN 61000-4-11 and EN 61000-4-29

1.5.5. Firmware upgrade

Mandatory - The Utility shall be able to update / change the meter firmware locally via the optical port or other ports installed on the meter.

The firmware upgrade mechanism shall be designed so as not to alter in any way the metering characteristics (metrology) or the meter data stored in the meter:

1. Meter data
2. Meter status
3. Configuration parameters or operational parameters

All of these data must remain unchanged even after firmware upgrades.

New firmware will be submitted to the meter with date and time parameter of executing new firmware operation (the meter will upload the new updated firmware but will only execute it when the defined date and time parameter achieved).

The meter will record the date and time of receipt of new firmware in the event log, as well as the date and



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time of the firmware upgrade process execution.

The meter will perform self-check process after the execution of the new firmware update, and the result of self-check process will be stored on the meter event log, and shall be retrievable locally or remotely.

The meter shall have non-volatile memory equal to or greater than ten times the size of the operational/running firmware image.

The meter shall support two concurrent firmware images: current operational/running and pending upgrade

1.5.6. Security

The meters shall use a method for encryption/decryption of data exchanged in all meter interfaces. According to cyber security department penetration test.

Mandatory - A method employing minimally AES128 shall be used in all information exchanges in private network, between meter and RS gateway (Field Area Network, or FAN interface), and between the meter and hand held unit (HHU) via the optical interface. Enough reserve capacity in the components of the system (e.g., main processor, memory) shall exist to support AES256 during the service life of the equipment.

The meter shall support encryption and decryption of two meter firmware images: current operational/running, pending upgrade.

The meter should be capable of supporting AES256 during its operational lifetime without a hardware change.

A method employing at least ECC192 shall be used in all information exchanges on public network, as well as between meter and central system directly.

The security methods and hardware shall be described in details by bidder.

The bidder shall prove following items:

1. Security of method
2. Client ability to change security key remotely
3. Confidentiality of remote key changing
4. If client changes the security key there is no way for meter manufacturers to retrieve new key (new key is not accessible for meter hardware, firmware, and external software)
5. All input and output information decrypted and encrypted

1.5.7. Meter Operational Modes

The meter shall be required to support at least two modes of operation: Normal Mode and Test Mode.

1.5.8. Meter Test Mode

The meter shall support a Test Mode that is activated by using the optical port.

In Test Mode, there shall be an automated test sequence available that includes:



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1. Full diagnostic test, that testing of the all the active and inactive functionality
2. Metering accuracy test (enabled to allow for accuracy testing to be performed without affecting the recorded customer registration for energy)
3. Connection validation tests (three-phase sequence connection).

The Test Mode shall also have the following displayed information (scrolled sequence):

1. Meter firmware version
2. Meter firmware update date and time stamp
3. Meter tampering events
4. Current limit
5. Connect/ disconnect

1.5.9. Load Profile

1. At least four programmable channels for:
 - a. Active interval energy (import and export)
 - b. Reactive interval energy (import and export) *
 - c. Active cumulative energy (import and export)
 - d. Reactive cumulative energy (import and export) *

Adjustable time interval from 1 to 60 minutes.
Meter shall store these parameters for at least 30 days.
2. At least four programmable channels for voltage and current as well as active and reactive power. Measurement type for each channel can be chosen as follows:
 - a. average
 - b. minimum
 - c. maximum

Adjustable time interval from 1 to 60 minutes, Meter shall store these parameters for at least 30 days.
3. Time & date shall be included in load profile.
4. Load profile shall be readable in complete form and in defined blocks (based on start and end time and channels) locally and remotely.



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1.5.10. Historic Consumption

For this option, there are twelve energy registers that store the total energy consumed for the previous twelve months. These registers are cycled at during the month-end process, with the oldest data being deleted.

1.5.11. Date and time management

The meter shall be equipped with a real time clock supporting the date and time. Pseudo-clock using mains crossing detection is not acceptable. The maximum drift of the clock shall be less than 30 seconds per month. The battery shall operate maintenance-free for a period of at least 10 years.

The meter shall be equipped with a battery monitoring system that monitors the battery's condition and sends a "Change Battery Alarm" message to the HES when the battery approaches the end of its operating life

When first commissioned into the AMI System the meter shall synchronize its time & date with that of the HES. Meter time & date synchronization shall also be done on a periodic basis through the HES, with a programmable periodicity, and done automatically during meter installation stage.

1.5.12. Power quality

1. Meter shall record under voltage, overvoltage, and power cut.
2. Events related to under voltage and overvoltage shall be recorded in the meter. Threshold of under voltage is from 110 to 66 by 5V increments and for overvoltage is from 110 to 143 by 5V increments
3. The voltage events will not be recorded unless they continue for a time equal to or greater than the programmable time set for under voltage and overvoltage threshold. This time shall be adjustable / programmable between 1 to 60 seconds, in 1-second increments.
4. For each under voltage, minimum voltage occurred during the adjustable / Programmable (1 to 60) Second period, the meter shall be detected and recorded as such; date and time of last occurrence and the cumulative number of the occurrences.
5. For each overvoltage, max. Voltage occurred and during the adjustable / Programmable (1 to 60) Second period the meter shall be detected and recorded as such; date and time of last occurrence and the cumulative number of the occurrences.
6. Parameters related to threshold and duration of under voltage and overvoltage shall be adjustable locally and remotely.
7. Events related to power cut shall be recorded in the meter as follows:
 - a. For long duration power cuts (more than 3 min.), duration time and date shall be recorded.
 - b. For short duration power cuts, number of power cuts shall be recorded.

1.5.13. Meter Display

1. Meter display shall display all measured quantities or desired quantity automatically or scrolling via button on the meter as follows:



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- a. Active energy in each quadrant
- b. Reactive energy in each quadrant
- c. Active energy in first & fourth quadrant
- d. Reactive energy in first & second quadrant
- e. Current Max. Demand and the Max. Demand with exact time & date monthly & quarterly
- f. Time and date
- g. Serial number
- h. Battery failure
- i. The number of voltage cut offs and number of reverse energy events
- j. Cell code no
- k. CT. & VT ratio
- l. average power factor (monthly)
2. Meter shall have capability of display and record of Max. Demand with ability of configuration of Max. Demand time interval calculation. Max. Demand shall be measured at least in three integer and two decimal digits.
3. LCD shall display measured items in OBIS codes.
4. Meter Display shall support Arabic language Interface).

1.5.14. Fault & Status Display

The meter shall provide a visible indication of the status of the incoming supply.

1.5.15. Status/Alarm/Event Displays

In addition to the displays required above, the meter must display the following minimum information in an intuitive way such that an inexperienced user can understand and interpret the information intelligibly (laminated instruction cards are to be provided for each unit):

1. Meter ID
2. Meter tamper state
3. Meter failure and/or fault code

The bidder must submit a detailed specification of the way in which the displays are structured and laid out.



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2. Electrical Requirement

The meters supplied shall meet the following electrical requirements.

Table (7)

No	Description	Specification
1	Meter Type	3-phase, 3-wire/4 wire, transformer-operated (CT and PT) meter, active & reactive energy, four quadrant measurements
2	Base current, I_b	CT Operated: 1A
3	Maximum continuous current, I_{max}	CT Operated: 6A
4	Starting current	0.1% I_b according to IEC 62053-22
5	Class index	Class 0.2 for Active Energy according to IEC 62053-22 Class 2 for Reactive Energy
6	Number of display digits	10
7	Display parameters Note: units on active energy (Wh), reactive energy (VARh), and demand (W) shall be configurable as Kilo or Mega	i. Active Energy for the current billing period ii. Reactive Energy for the current billing period iii. Cumulative Total Active Energy used since installation iv. Cumulative Total Reactive Energy used since Installation v. Last month consumption vi. Date and time vii. Instantaneous primary voltage for each phase in V viii. Instantaneous primary current for each phase in A ix. Instantaneous PF for each phase x. Maximum Demand xi. CT Ratio xii. PT Ratio xiii. transformer code no.

3. Meter Terminal Block Specifications

- The Meter Terminal Block shall be made from fire-resistant material (polycarbonate grade 500R or equivalent Bakelite) materials.
- The meter Terminal Block cover shall be made from fire resistance material (polycarbonate), and shall be extended type.
- The meter terminals shall have two flat-head screws for wiring connection, these screws shall be made of brass. Furthermore, the meters shall meet the following terminal requirements

Table (8)

No	Description	Three Phase Direct Connected
1	Terminals Materials	brass or copper current terminals
2	Terminal bore Diameter	3 mm



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4. event detection

The meter shall classify and record the following commonly-occurring tamper events with a date and time stamp. The tampering event log shall record any combination of the following possible tamper conditions:

Table (9)

Event Case	MV meter
Current Reverse	Phase current reverse (L1, L2, L3) in separate or combination.
Current imbalance	$I_1 + I_2 + I_3 \neq I_n$
	Neutral current =0
Voltage Missing	For any phase (L1, L2, L3).
Current missing	For any phase (L1, L2, L3).
External magnetic field influence	√
Opening meter cover	√
Opening terminal cover	√

- 4.1. There shall be a mechanism by which the consumer can be disconnected immediately when a verifiable tamper event occurs.
- 4.2. The event status shall be reset by using the meter optical port.
- 4.3. The meter shall send a push message to the HES with every Event, stamped with date and time.
- 4.4. The meter shall provide EVENT log data with the last occurred EVENT date and time stamp and it can be easily readable remotely or locally.



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5. Meter Metrology

1. The meter shall have the ability of measuring and recording import/export active energy

(The values to be recorded for import and export are actual values).

2. The meter should have the ability of measuring and recording import/export reactive Energy (the values to be recorded for import and export are actual values).

3. The meter shall measure phase current (True RMS) in A

4. The meter shall measure phase voltage (True RMS) in V

5. The meter shall measure Power Factor

6. The meter shall measure Load profile

7. The measuring quantity for energy shall have following resolutions:

- a. for Active Energy (Min 0.01 kWh resolution).

- b. for Reactive Energy (Min 0.01 kVARh resolution)

8. It shall be possible to locally and remotely select or configure exported energy whether to be registered or not.(according to standard, when the energy flows from network to the consumer it is registered as IMPORT and when the energy flows from consumer to the network it is registered as EXPORT.)

9. The meter is configured in such way not to register the consumed energy as Export (except Distributed Generations) and whenever it is occurred, the system shall record this event.

10. The meter shall measure import/export energy as well as Absolute value of active energy and it shall be possible to activate this capability.

11. The measuring and recording maximum demand of active power shall be based on configurable time intervals and subintervals (sliding mode).

- a. It shall be possible to reset maximum demand remotely

- b. It shall be possible to reset maximum demand locally via sealed push button on front of the meter.

- c. It shall be possible automatically Maximum demand to be reset at specific billing Period.

Above capability shall be activated (or not) and all parameter shall be adjustable remotely and locally.

12. The meter shall have the ability to measure the angle between voltage phases

13. The meter shall have the ability to measure Maximum Demand

14. The meter shall have the ability to give phasor diagram for the measured voltage & current (monitor & save it to meter computer interface program and/or HES)

15. Note: units on active energy (Wh), reactive energy (VARh), and demand (W) shall be configurable as kilo or Mega

6. Tariff (For Private Feeders /transformer)

- 6.1.1. All tariff calculation and credit calculation shall be processed inside the meter.

- 6.1.2. The meter shall support stepped tariffs structures as well as time-of-use and maximum demand.

- 6.1.3. The tariff structure consists of the following records:

1. Each tariff must be uniquely identified using the tariff code

2. Each tariff shall have an activation date, being the date on which the tariff becomes



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effective.

- Each tariff has up to ten steps / time slot (in kWh) for different levels of energy pricing.
- Each tariff has up to ten steps / time slot (in kVARh) for different levels of energy pricing.
- The rate describes the cost per kWh for energy consumption in that slot.
- The rate describes the cost per kVARh for energy consumption in that slot.

6.1.4. Tariff switchover

When the meter detects that a new tariff is applicable (using the tariff activation date), the meter shall execute the following steps:

- The meter shall begin the billing against the new tariff.
- The current tariff code shall be updated to reflect the new tariff code.
- The old (expired) tariff should be deleted.
- New tariffs will only be activated at 00:00 on the first day of a month.

7. Billing

Billing operates on a “monthly” basis (a period of days definable by the Utility). The energy consumption charge is deducted within the meter for at least every 1 kWh. All other charges including service charge, minimum charge, VAT, and other taxes are deducted at the POS.

8. Smart Meters - Testing and Approval

8.1. Routine Acceptance Tests

- Routine / Acceptance tests shall be carried out on all offered meters, and shall be carried out by Manufacturer / Contractor, with providing a Factory acceptance test certificate for the manufactured meters.
- The tests shall be carried out in accordance with IEC 62053-21, IEC 62053-22, IEC 62052-11, BS EN50470-1, BS EN 50470 – 3, BS EN 61268, and Tender Requirements.
- Number of Samples Selection Criteria Done in accordance to the IEC 514 Standard. (**)



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Table (10)

No.	Test Name			As per Standard
1	Test of Meter Constant			As per IEC 62052-11 , BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
2	Limits of error due to variation of the Current			As per IEC 62052-11, BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
3	Test of no-load condition			As per IEC 62052-11, BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
4	Test of starting condition			As per IEC 62052-11, BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
5	Voltage Variation			As per IEC 62052-11 , BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
6	Frequency Variation			As per IEC 62052-11 , BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
7	Test of power consumption			As per IEC 62052-11 , BS EN50470-1 and BS EN 50470 – 3 for active , BS EN 61268 for reactive
8	Influence of Self-Heating Test(*)			BS EN 50470 – 3 for active , BS EN 61268 for reactive
9	Influence of Ambient Temperature Variation(*)			BS EN 50470 – 3 for active , BS EN 61268 for reactive
10	Influence of short time overcurrent(*) (if applicable)			BS EN 50470 – 3 for active , BS EN 61268 for reactive
11	Resistance to heat and fire test(*)			BS EN 50470 – 3 for active , BS EN 61268 for reactive
12	Visual Inspection			
		1	Meter Display	As per RFP Specification
		2	Meter Casing	As per RFP Specification
		3	Meter Sealing	As per RFP Specification
		4	Meter name plate Marking	As per RFP Specification
		5	Connection Diagram and terminal Marking	As per RFP Specification
		6	Packing	As per RFP Specification
13	event Test			
	Three Phase Tests	1	Phase (I1 , I2 , I3)Current and or Neutral Current Reverse Connection	As per RFP Specification
		2	Current imbalance (I1+ I2 + I3 ≠ In)	
		3	Neutral Current = 0	
		4	Missing phase Voltage	
		5	External Magnetic influence	
		6	Open Meter Cover and or Terminal cover	
14	Functional tests			
		1	Meter Mode	As per RFP Specification
		2	Maximum Demand	As per RFP Specification
		3	Load Profile	As per RFP Specification
		4	Meter Local Communication (Optical , RS485)	As per RFP Specification
		5	Meter Remote Communication	As per RFP Specification
		6	Tariff	As per RFP Specification
		7	Tariff Switchover	As per RFP Specification
		8	Load Control	As per RFP Specification



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(*) The first random sample will be collected from the first delivered batch, and the other random sample will be collected from the rest of the delivered meters.

In case of Sample test failure the delivered meter batch will be rejected, then collect other random sample from each later delivered batches in order to apply these tests , and the order will completely canceled in case of the failure of applying these tests three times.

(**) Number of Samples Selection Criteria according to IEC514

Table (11)

No. of Meters per batch (N)		First Sample			Second Sample	No of Meters for both 2 samples (n1+n2)
		No of meters per sample (n1)	No of Rejected meters (d1)	No of Allowed rejected meters (c1)	No of meters per sample (n2)	Total No of Allowed rejected meters(C2)
1 st Case	101-500	30	2	0	30	1
2nd Case	501-1000	40	2	0	40	2

8.2. Meter Type test (mandatory)

The meter shall have a type test certificate that include details of test results according to IEC62052-11, IEC62053-21, IEC62053-22, IEC 62053-23 , BS EN 50470 – 3, and BS EN 61268 from international accredited Laboratory (KEMA, NMI Holland, CESI Italy, or any worldwide recognized lab).

Mandatory - The bidder shall provide a type test report issued within the last 5 years. (Submission is Mandatory before the Pilot) The type test must include all tests that applicable that particular type in the standards. Minimally expected tests are:



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Table (12)

NO	Test Name	As per Standard
1	Test of insulation properties	
1.1	Impulse voltage tests	IEC62052-11 ,EN50470-1
1.2	AC Voltage tests	IEC62052-11,EN50470-1
2	Tests of accuracy requirements	
2.1	Test of meter Constant	BS EN 50470 – 3 for active , BS EN 61268 for reactive
2.2	Test of Starting Condition	BS EN 50470 – 3 for active , BS EN 61268 for reactive
2.3	Test of No-Load condition	BS EN 50470 – 3 for active , BS EN 61268 for reactive
2.4	Test of influence quantities	BS EN 50470 – 3 for active , BS EN 61268 for reactive
3	Test of Electrical Requirements	
3.1	Test of power consumption	BS EN 50470 – 3
3.2	Test of influence of supply voltage	IEC62052-11 , EN50470-1
3.3	Test of influence of short-time overcurrent	BS EN 50470 – 3
3.4	Test of influence of self-heating	BS EN 50470 – 3
3.5	Test of influence of heating	IEC62052-11 , EN50470-1
3.6	Test of immunity to earth fault	IEC62052-11 , EN50470-1
3.7	Test of accuracy in the presence of harmonics	EN 50470-3
4	Test for Electromagnetic Compatibility (EMC)	
4.1	Test of immunity to electrostatic discharges	IEC62052-11 , EN50470-1
4.2	Test of immunity to electromagnetic RF fields	IEC62052-11 , EN50470-1
4.3	Fast transient burst test	IEC62052-11 , EN50470-1
4.4	Test of immunity to conducted disturbances, induced by radio frequency	IEC62052-11 ,EN50470-1
4.5	Surge immunity test	IEC62052-11 ,EN50470-1
4.6	Damped oscillatory waves immunity test	IEC62052-11 , EN50470-1
4.7	Radio interference suppression	IEC62052-11 ,EN50470-1
5	Test of effect of the climate environments	
5.1	Dry Heat test	IEC62052-11 , EN50470-1
5.2	Cold test	IEC62052-11 ,EN50470-1
5.3	Damp heat cyclic test	IEC62052-11 ,EN50470-1
5.4	Solar radiation test	IEC62052-11 ,EN50470-1
6	Mechanical Test	
6.1	Vibration test	IEC62052-11 ,EN50470-1
6.2	Shock test	IEC62052-11 ,EN50470-1
6.3	Spring hammer test	IEC62052-11 ,EN50470-1
6.4	Tests of protection against penetration of dust and water	IEC62052-11 ,EN50470-1
6.5	Test of resistance to heat and fire	IEC62052-11 , EN50470-1



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8.3. DLMS Compliance Certificate (mandatory)

Mandatory - The meter shall have a certificate for IEC62056 DLMS/COSEM, for the following:
(Submission is mandatory upon submission of proposals)

1. Mandatory - Optical port Communication (Submission is mandatory upon submission of proposals)
2. Mandatory - 2-wire Serial RS485 Communication port (Submission is mandatory upon submission of proposals)
3. Mandatory - WAN / FAN Communication Modem (Submission is mandatory upon submission of proposals) failure to submit any one of the theses Certificates will be considered as the Bidder failing to submit the whole item (DLMS Compliance Certificate).

8.4. Communication Modem Certificate

Mandatory - Communication modem should have certificate for industrial equipment from an international accredited laboratory (Submission is mandatory before the Proof of Concept)

8.5. Meter Life Time Test Certificate (mandatory)

Mandatory - The meter must be certified for 15 to 20 years life of operation, according to the Accelerated Life Time (ALT) IEC testing Procedures found in IEC62059-41, IEC62059-31-1, and IEC62059-32-1. (Submission is mandatory before the Proof of Concept)



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9. Connections diagram:

For Distribution point (Distributers) , All smart meters in distributer cells is connected to RS gateway via RS485 port as shown in figure below and RS gateway transmit connected meters data to HES via 4G (at least) communication .

For Kiosks, Telecommunication Technology is 4G (at least).

